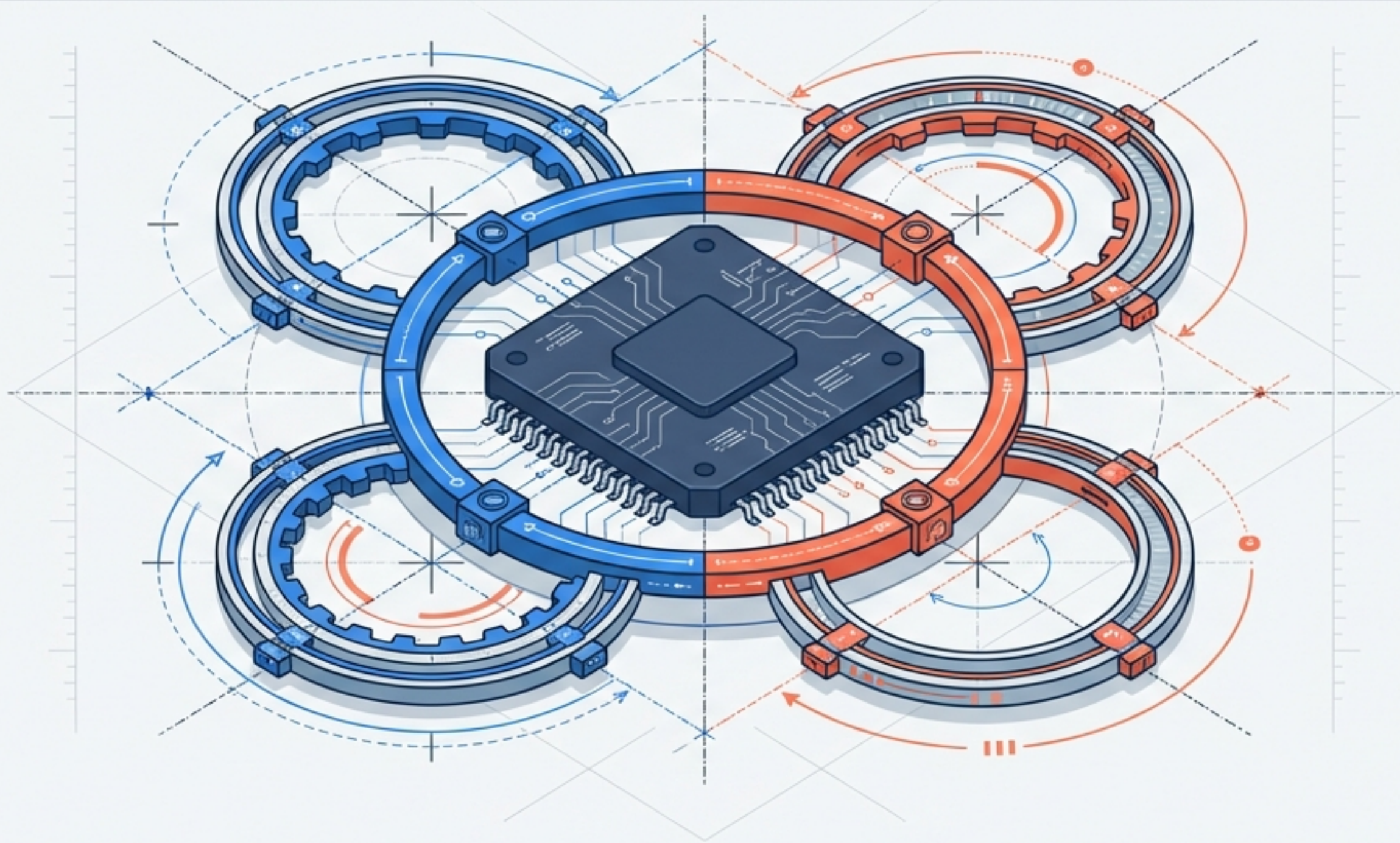


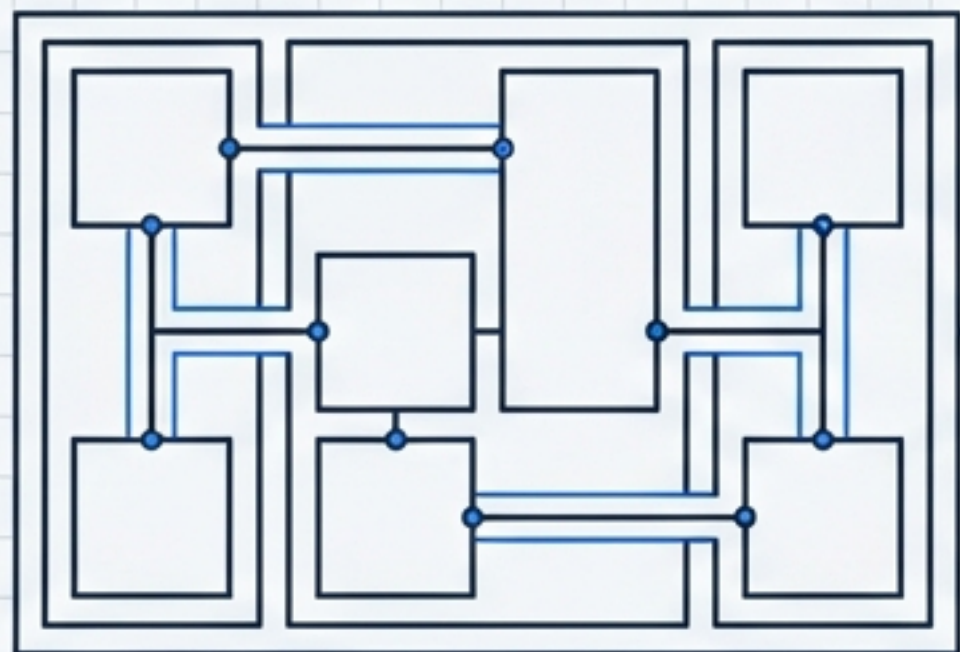


AIエージェントの制御アーキテクチャ：ハーネスエンジニアリング

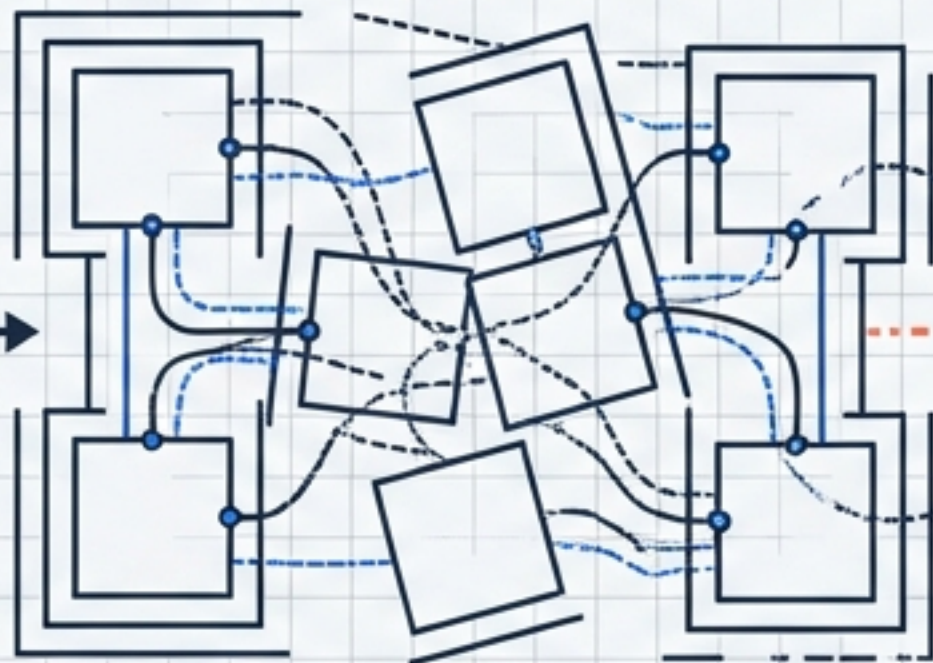
「お願い」から「構造的制約」へ。B2B SaaS開発におけるLLMの自律制御と運用基盤

2026年4月
SaaS Engineering & Architecture

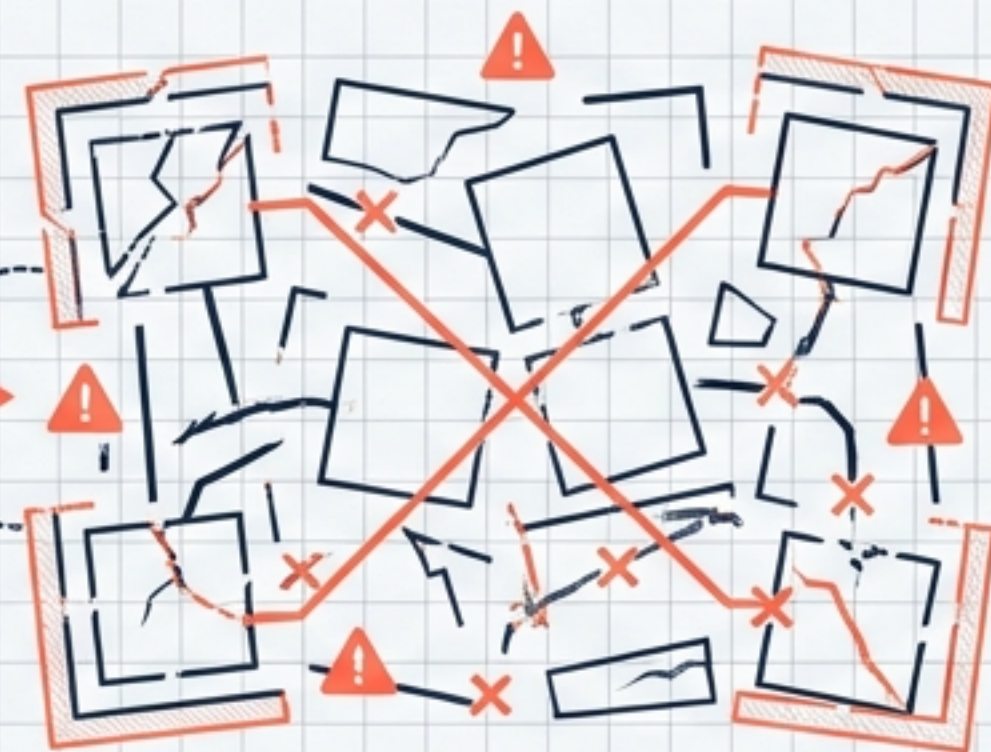
「Vibe Coding」の限界：なぜプロンプトだけでは破綻するのか？



Day 1: セッション開始



Session 2: 文脈の消失



Session 3: 構造の崩壊



1. 揮発するコンテキスト

昨日のAIは今日の文脈を知らない。セッション間で仕様や設計思想がリセットされ、「Lost in the Middle」現象に陥る。



2. チーム一貫性の崩壊

各開発者がパラパラのプロンプトで指示を出すため、コードベース全体でのアーキテクチャや命名規則の統一が維持できない。



3. 監査・保守のブラックボックス化

なぜその実装になったのか、意図（Intent）が個人のチャット履歴にしか残らず、B2B SaaSに必須の監査に耐えない。

パラダイムシフト：AI制御の進化段階

「お願い」から「構造的制約」へのシフト

Stage 1: Prompt Engineering (2023~)

対象: 単一の応答
(Single Response)

制御方法: テキスト指示
(Text Instructions)

違反検知: なし
(None)

寿命: セッション内のみ

Stage 2: Context Engineering (2024~)

対象: 単一セッション
(Single Session)

制御方法: ファイル配置 /
CLAUDE.md

違反検知: 手動レビュー
(Manual)

寿命: タスク完了まで

Stage 3: Harness Engineering (2025~)

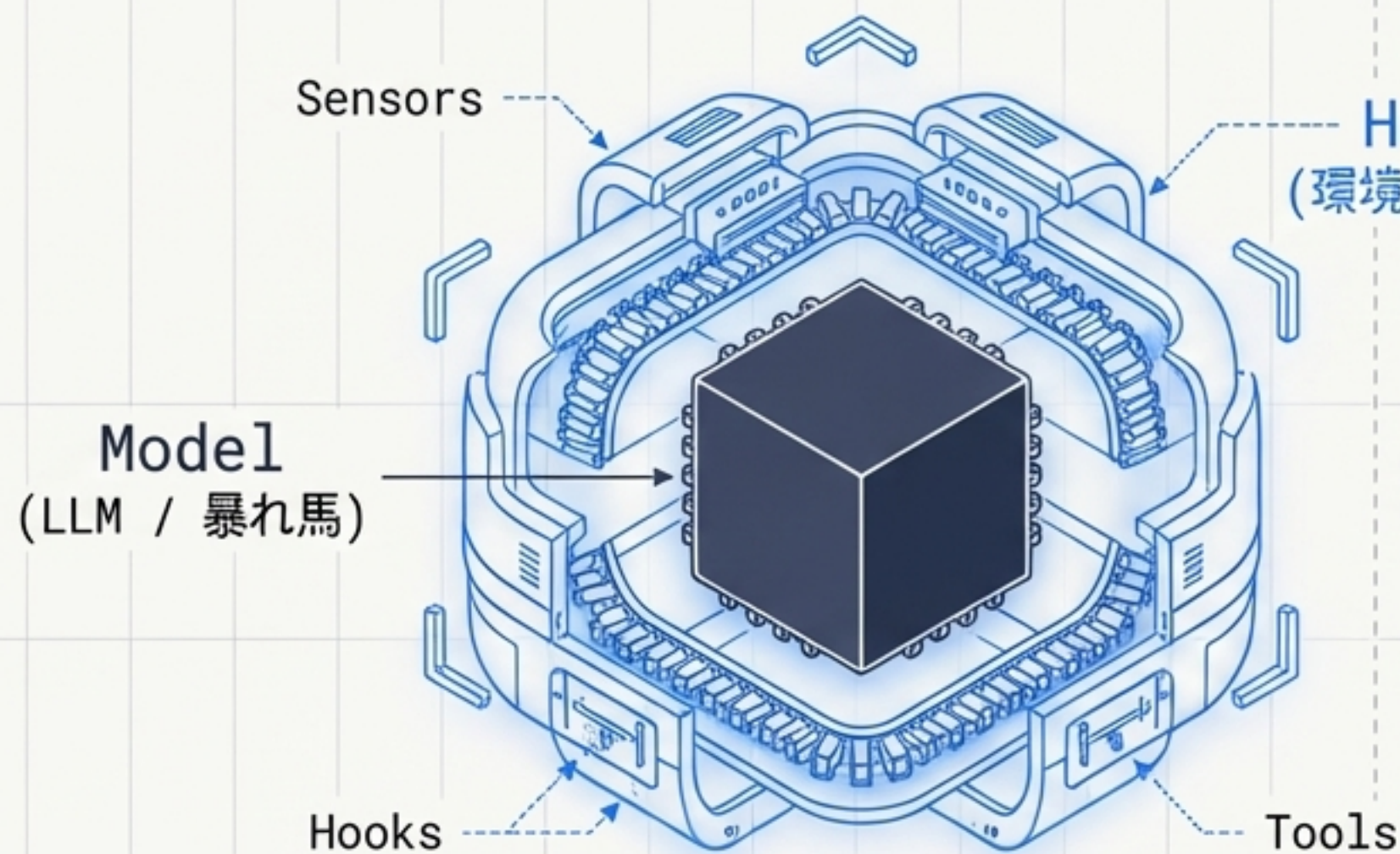
対象: 複数セッション
(Multi-Session)

制御方法: 構造的制約・自動フック
(Constraints)

違反検知: 即時・自動
(Immediate/Auto) ⚠

寿命: プロダクトの寿命

Agent = Model + Harness



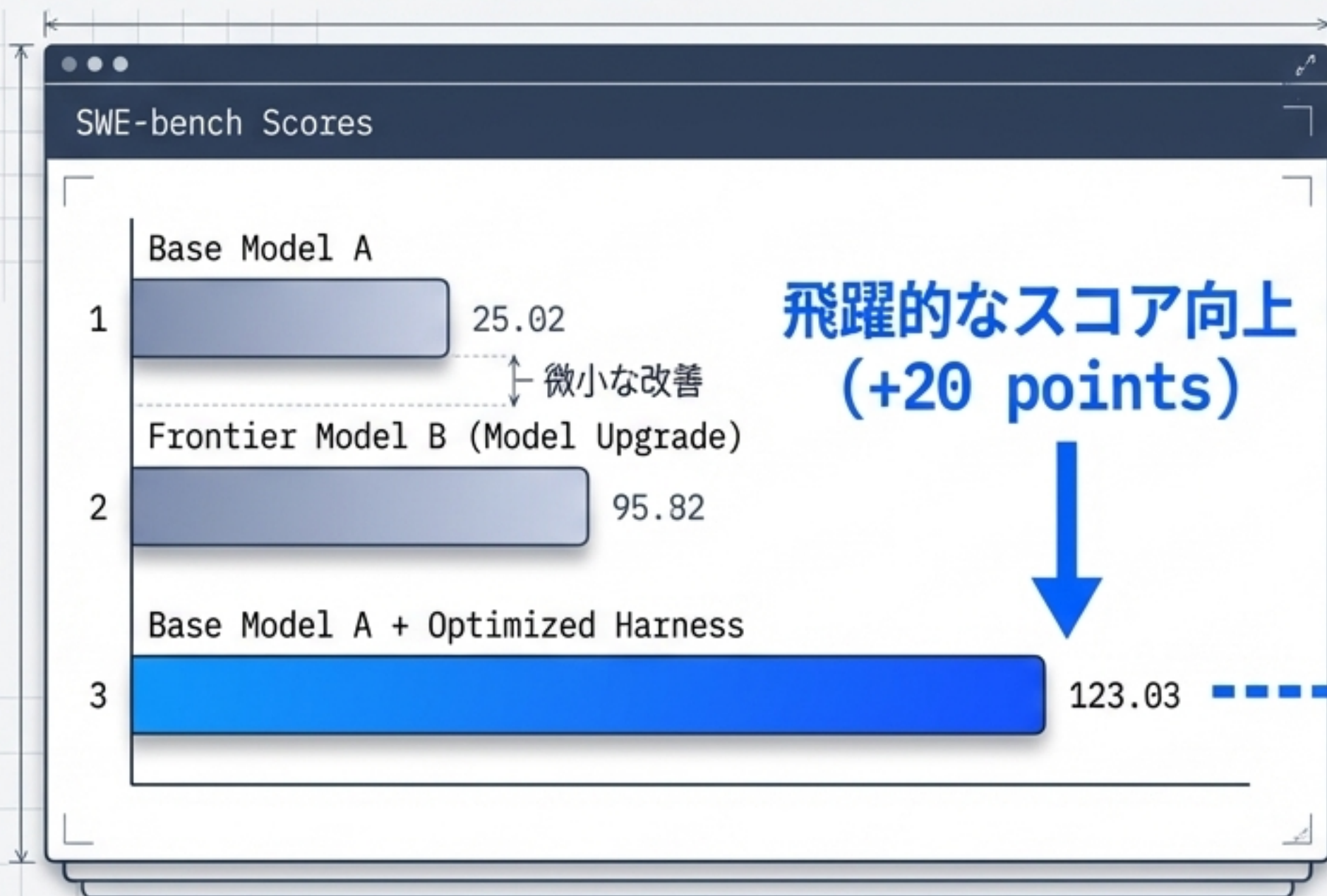
$$= \text{Model} + \text{Harness}$$

ハーネス（馬具）とは、暴れ馬（LLM）の脚力を削ぐものではなく、最大限の力を正しい方向へ導くための環境設計。

実績：OpenAI Codexチームは、手動コード一切なしで5ヶ月間に100万行超のコードを出荷。

Humans steer. Agents execute.
(人間が舵を取り、エージェントが実行する)
— Ryan Lopopolo, OpenAI

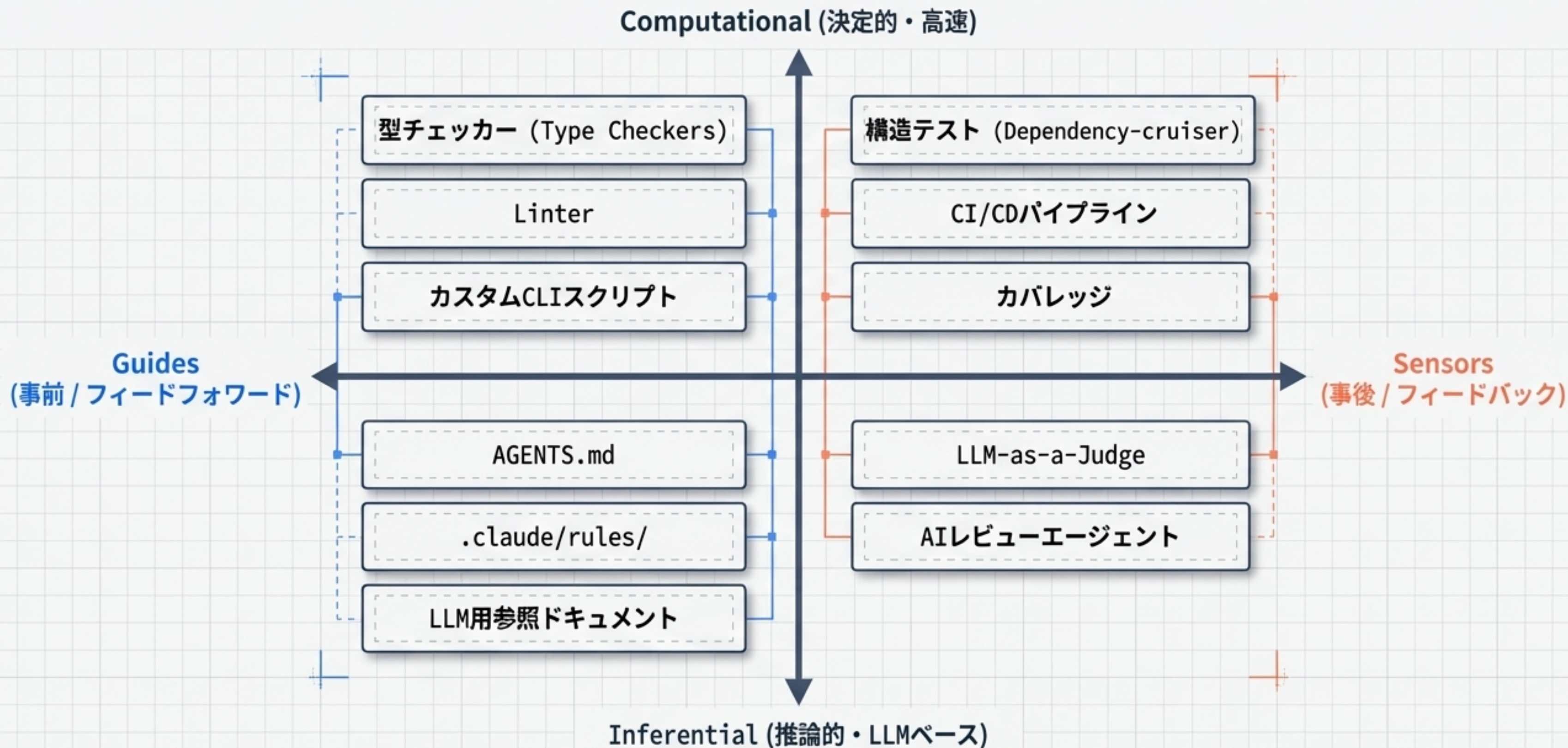
モデルのアップデートを待つより、 環境（ハーネス）を構築せよ



Data Insights

- SWE-benchのデータが示す事実：
同一モデルであっても、最適なスキ
ャフォールド（足場）とフィードバ
ックループを組むだけで、フロン
ティアモデルへの乗り換えを凌駕す
るスコア向上が見られる。
- エージェントの品質差は、「モデル
の性能差」ではなく「ハーネスの設
計力」で決まる時代へ。

サイバネティクス制御マトリクス：2軸で設計するガードレール



ハーネスを構成する5つのコア要素



要素解説①：コンテキストとルールの「段階的開示」

Anti-pattern

AGENTS.md

CONTEXT: This document contains information about the project and the agents that will be used to interact with it. It is intended to be a single source of truth for all agents, and should be kept up-to-date as the project evolves. The information in this document is organized into sections, each of which describes a different aspect of the project. The first section, "Project Overview", provides a high-level overview of the project and its goals. The second section, "Agents", describes the different agents that will be used to interact with the project, and the roles they will play. The third section, "Rules", describes the rules that will govern the interactions between the agents and the project. The fourth section, "Guidelines", provides guidelines for how the agents should interact with the project. The fifth section, "FAQ", provides answers to common questions about the project. The sixth section, "References", provides links to other resources that may be useful for understanding the project. The seventh section, "Appendix", provides additional information about the project. The eighth section, "Index", provides a list of all the sections in the document. The ninth section, "Glossary", provides definitions for key terms used in the document. The tenth section, "Bibliography", provides a list of all the sources used in the document. The eleventh section, "Index", provides a list of all the sections in the document. The twelfth section, "Glossary", provides definitions for key terms used in the document. The thirteenth section, "Bibliography", provides a list of all the sources used in the document. The fourteenth section, "Index", provides a list of all the sections in the document. The fifteenth section, "Glossary", provides definitions for key terms used in the document. 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アンチパターン: 全ての仕様とルールを巨大な一つのファイルに詰め込む。Lost in the Middle現象により重要なルールが無視される。

Best Practice

AGENTS.md
(< 100 lines)

Md

.claude/rules/
architecture.md

Md

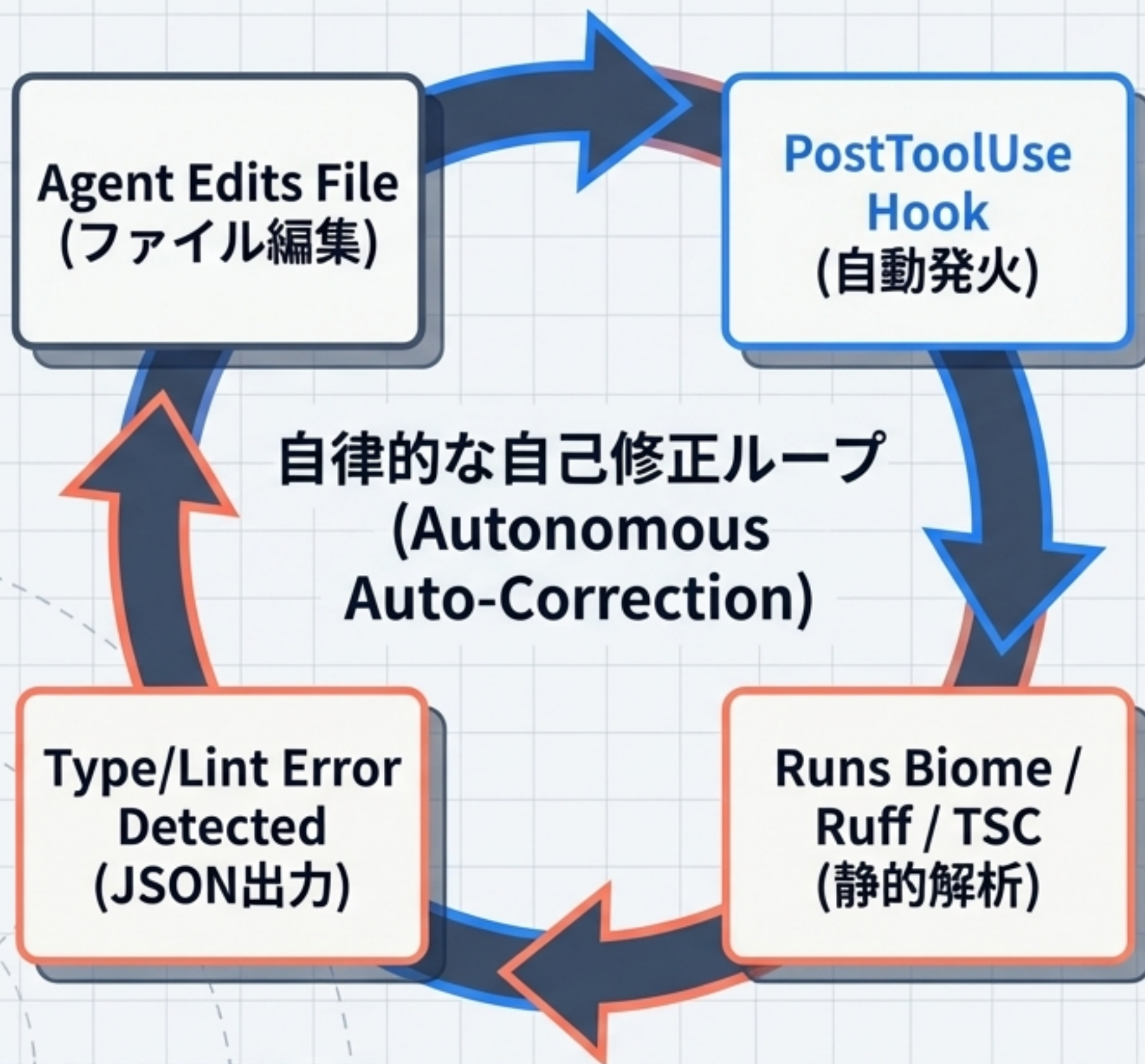
.claude/rules/
testing.md

Md

.claude/rules/
api-guidelines.md

ベストプラクティス: 段階的開示 (Progressive Disclosure)。目次のみを維持し、エージェントが必要な時にのみ特定ドメインのルールを動的に読み込ませる。

要素解説②：Hooksと高速フィードバックループ



- 人間の事後レビューを待たない。品質保証を左ヘシフト (Shift Left) する。
- エージェントがファイルを編集した瞬間に、計算的センサー (Linter、フォーマッタ) をHookで自動発火。
- エラーが出た場合は標準出力をJSONでエージェントに返し、瞬時に修正を促す。

Before / After : ハーネスが生み出す一貫性

Before: CLAUDE.mdのみ (Vibe Coding)

認証APIを追加して

src/auth.ts

src/user.ts

test/auth.test.ts

docs/auth_flow.md

config/auth_config.json

エラー多発: テスト欠如、フォルダ構造の不一致、
エラーハンドリングの欠落

After: ハーネス導入後

認証APIを追加して

src/auth.ts

src/user.ts

test/auth.test.ts

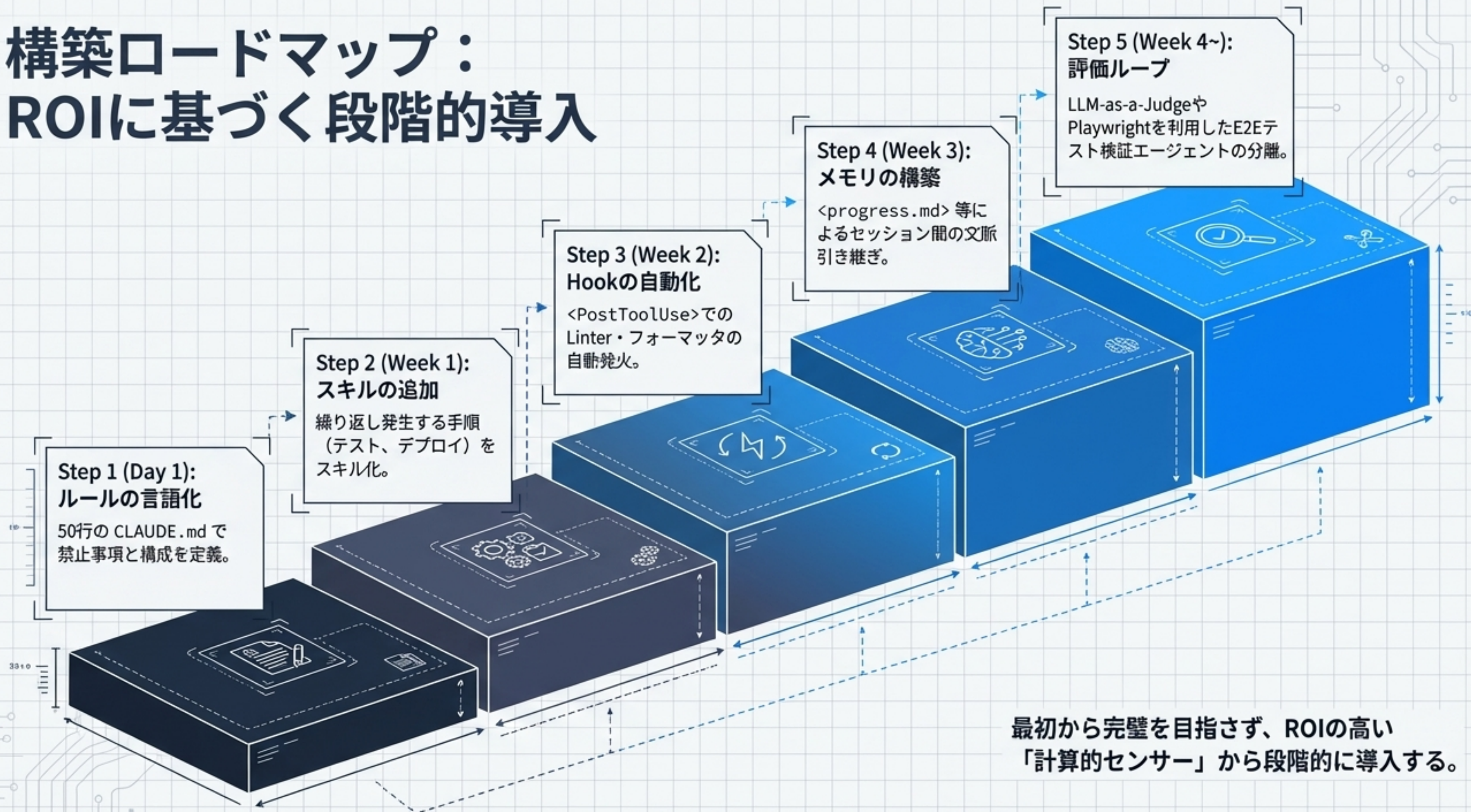
docs/auth_flow.md

config/auth_config.json

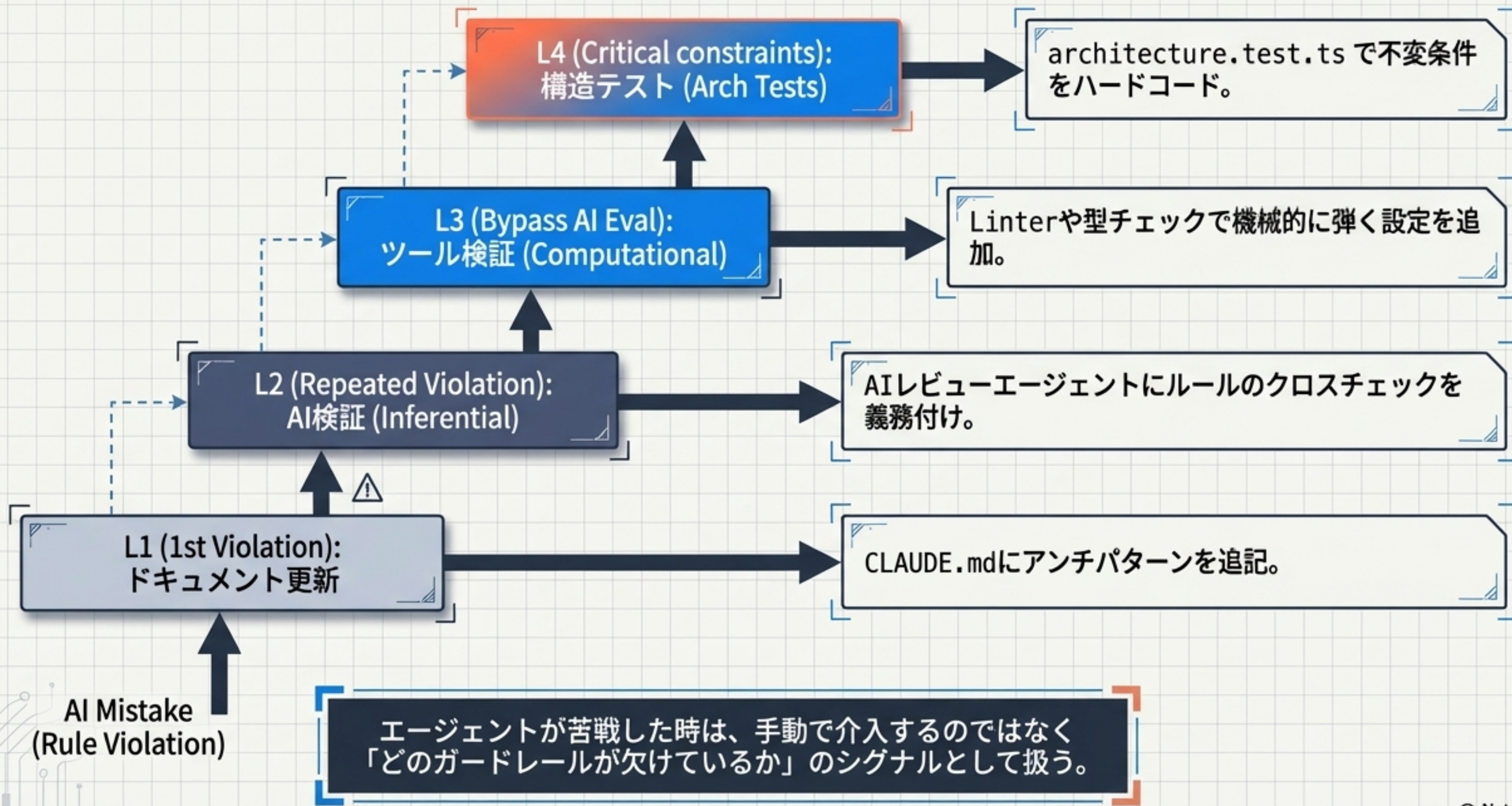
完璧な一貫性: テスト通過、正しいレイヤー配置、
ドキュメント自動生成

ハーネスなしでは「良い出力が出ることを祈る」運用。
ハーネスありでは「悪い出力が出にくい構造」が担保される。

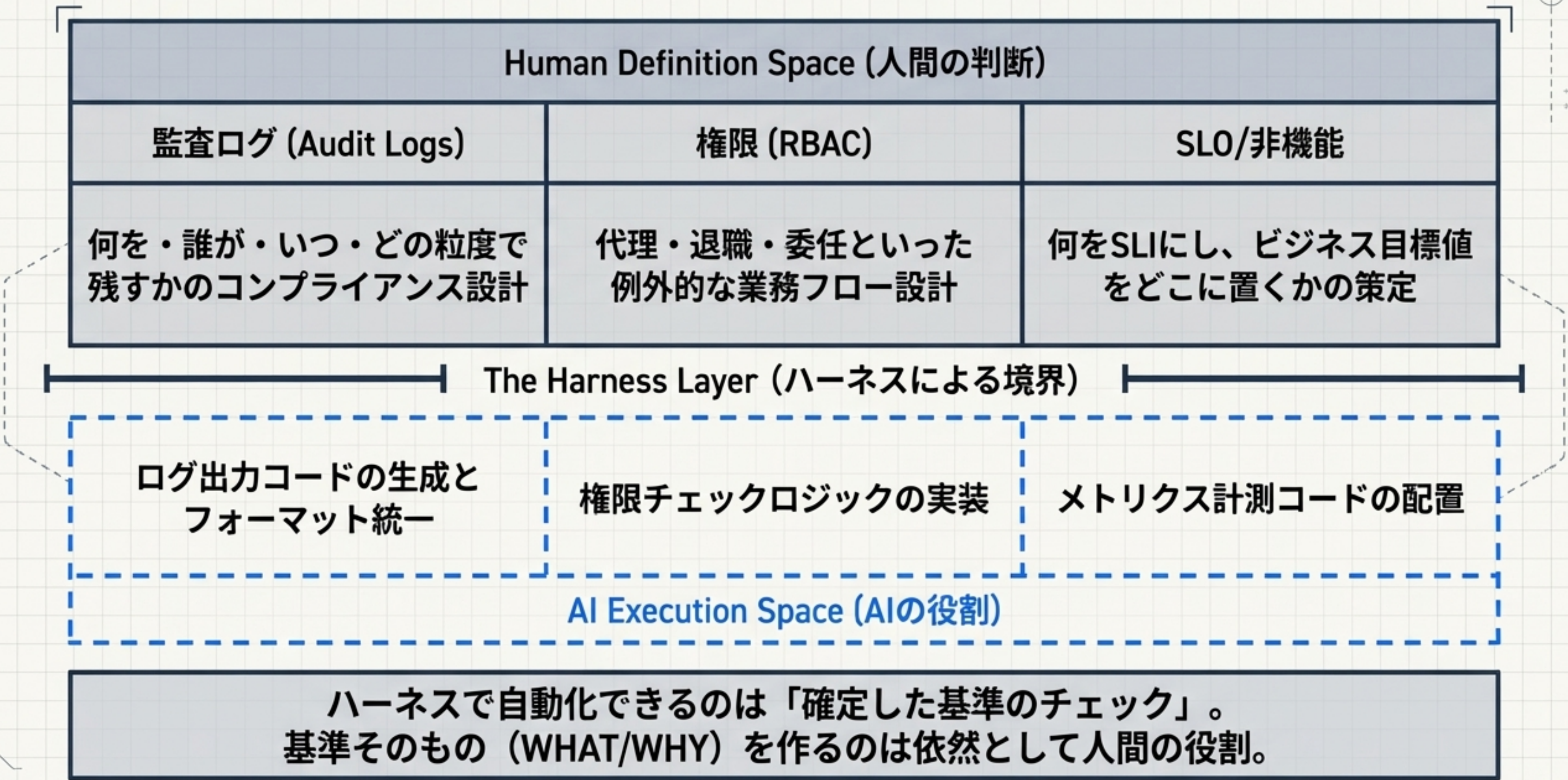
構築ロードマップ： ROIに基づく段階的導入



違反対応のエスカレーションラダー (3回ルール)

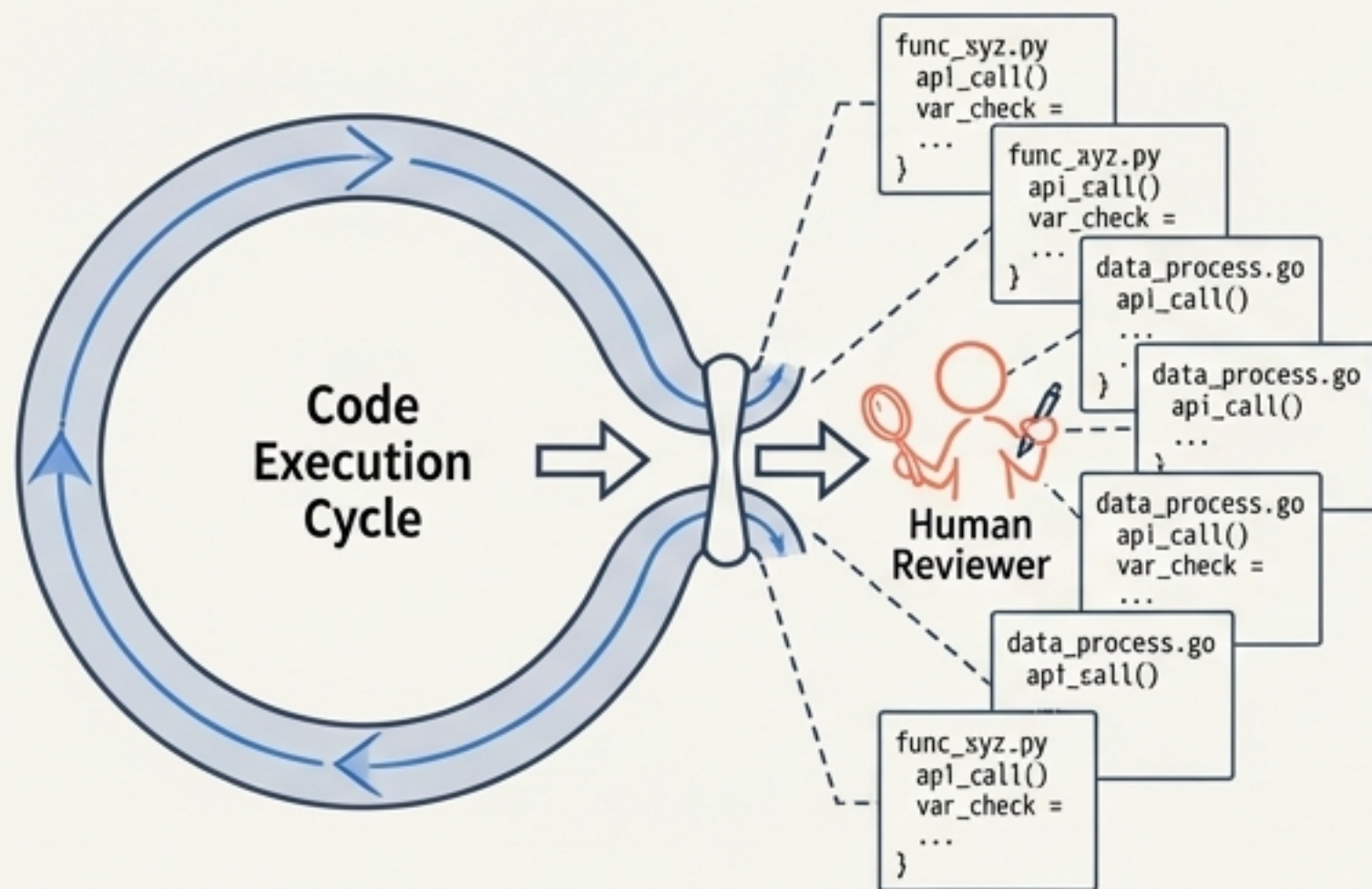


B2B SaaSにおける境界線：AIに任せられない「判断」



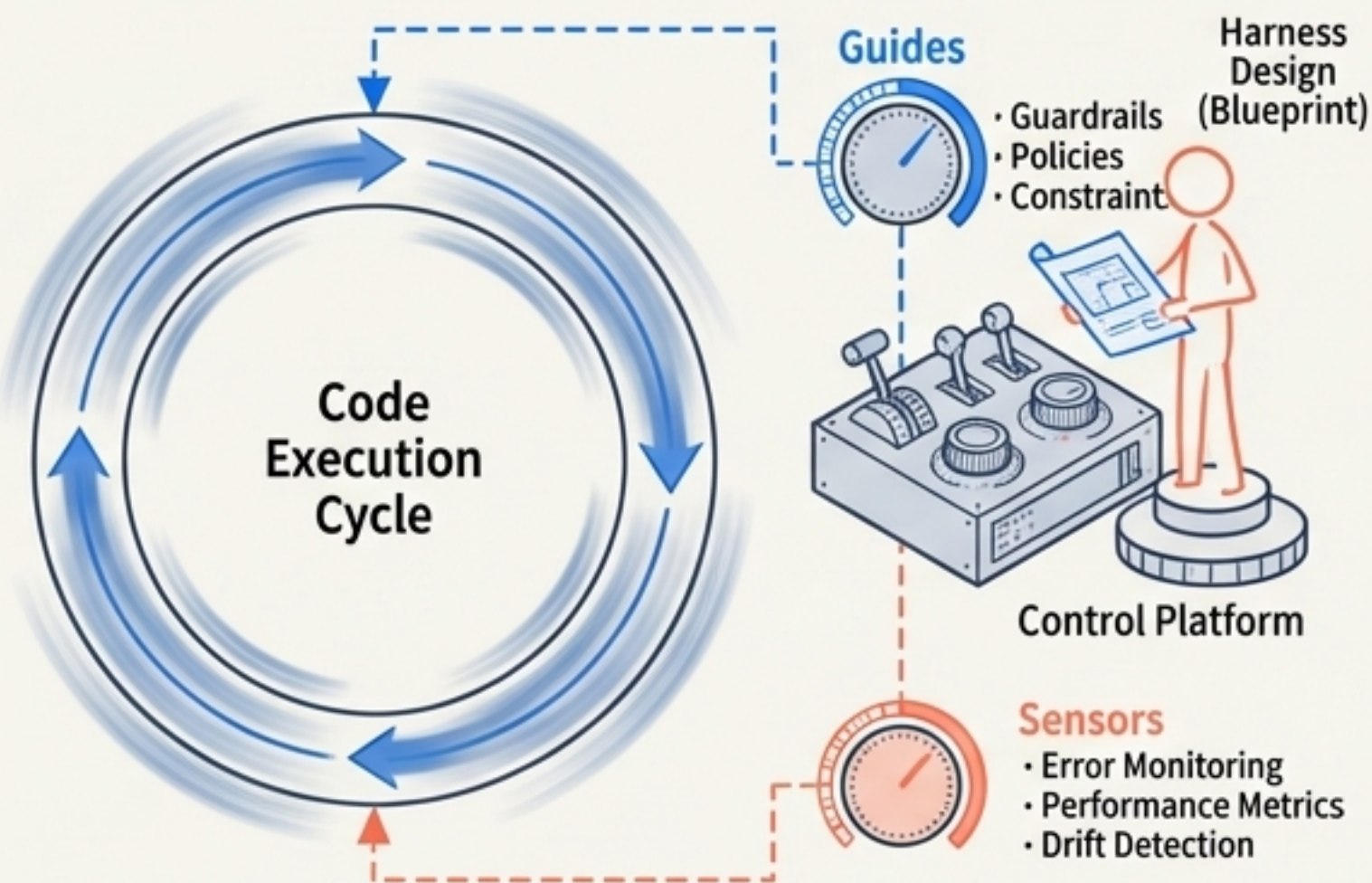
エンジニアの新しい役割：「Humans ON the loop」へ

Humans IN the loop (従来型)



人間がAIのコードを1行ずつレビューする（ボトルネック化）

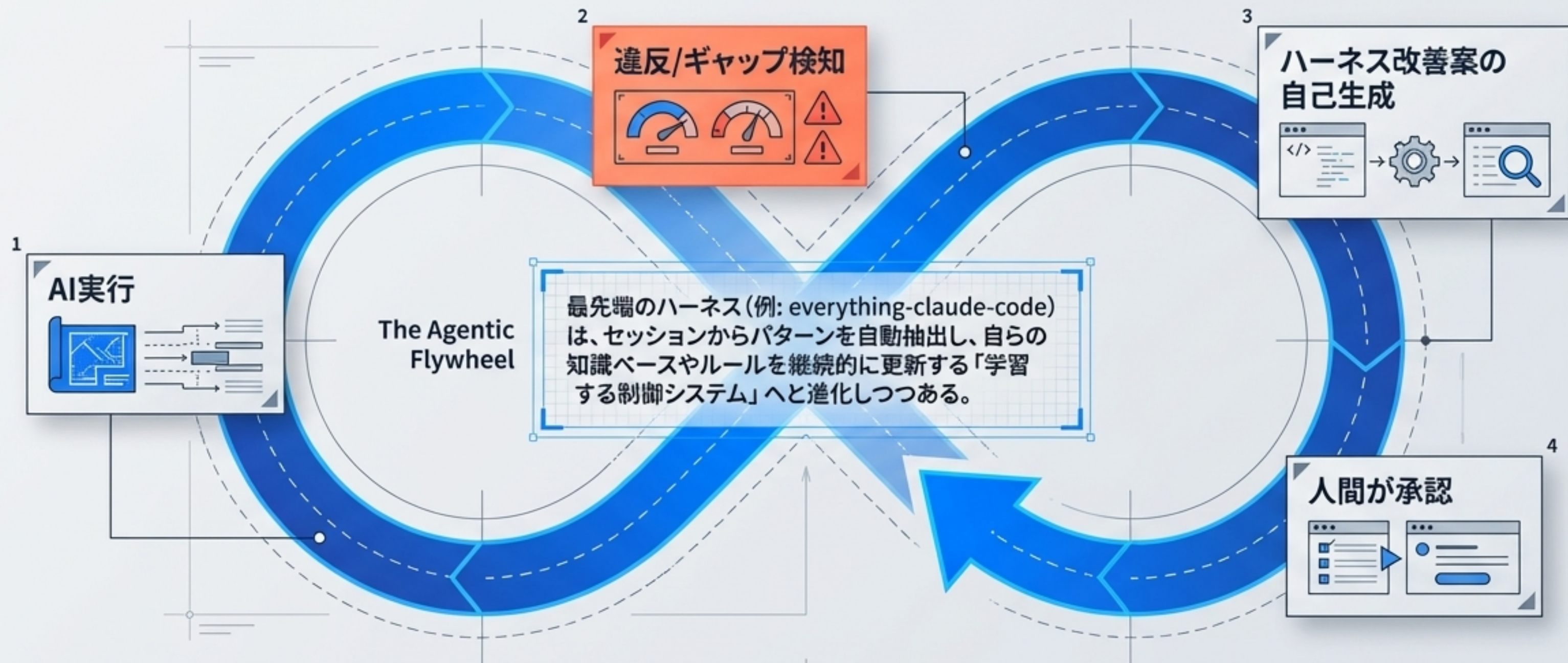
Humans ON the loop (ハーネスエンジニアリング)



人間はコードの記述から離れ、AIが正しく動作するための制御構造（ハーネス）の設計と改善に集中する

AIにコードを1行ずつレビューする存在（In the loop）から脱却する。
エンジニアの主戦場は「制御構造の設計者」へと一段上に移動する。

自己最適化するハーネスと、次世代の開発基盤



AIに「お願い」するのは、もう終わりにしよう。
エージェントが失敗できない環境を「エンジニアリング」する時代へ。